SAMSON JOSE J

CSE (Artificial Intelligence & Machine Learning) |AI/ML Enthusiast

Portfolio

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PERSONAL SUMMARY

Computer Science (AI & ML) undergraduate with hands-on experience developing machine learning and computer vision applications using Python, Scikit-learn, TensorFlow, PyTorch and OpenCV. Experienced in data preprocessing, model training, performance evaluation, real-time API integration and deploying automated data workflows. Developed academic and hackathon projects in medical image classification, air-quality forecasting and face anti-spoofing. Seeking an entry-level AI/ML opportunity to apply problem-solving skills and build practical, data-driven solutions.

EDUCATION

ATME College of Engineering, Mysuru **Sept 2023 - (2027)**

Bachelor of Engineering in CSE (AI & ML) | **CGPA: 8.87**

St. Philomena's PU College, Mysuru **Mar 2023**

PU in PCMC | **Percentage:91%**

St. Matthias High School, Mysuru **May2021**

KSEEB | **Percentage:73.6%**

TECHNICAL SKILLS

- **Programming**: Python, C++, C, SQL
- **Machine Learning**: Supervised Learning, Unsupervised Learning, Semisupervised Learning, Renforcement Learning , Classification, Regression, Clusters, Model Training, Model Evaluation
- **Libraries & Frameworks**: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, OpenCV, YOLOv8
- **Computer Vision:** Image Preprocessing, Image Classification, Face Detection, Face Alignment, Face Recognition
- **Data & APIs**: Data Preprocessing, REST APIs, API Integration, OpenWeather API, IQAir API
- **Tools & Platforms**: Jupyter Notebook, Google Colab, VS Code, Git, GitHub, Flask, Firebase, Render, GitHub Actions, Jira
- **Core Computer Science**: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Cloud Computing, SDLC

PROJECTS

Brain Tumor Recognition Using Machine Learning |Major Project

- Developed a four-class brain MRI classification system using 7,023 images covering Glioma, Meningioma, No Tumor, and Pituitary.
- Applied image preprocessing including 160×160 resizing, pixel normalization, and feature vectorization for ML model development.
- Implemented and compared KNN, MLP, Decision Tree, Random Forest, and Extra Trees classifiers using Scikit-learn.
- Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, with Extra Trees achieving 95.58% test accuracy.

Anti-Spoof AI Attendance System | 24-Hour Hackathon

- Developed an AI-powered attendance system using Python, Flask, OpenCV, PyTorch, and YOLOv8 for real-time face detection and anti-spoofing.
- Built a multi-stage computer vision pipeline using YOLOv8 for face detection, RetinaFace for face alignment, and AdaFace for 512-dimensional face embeddings and recognition.
- Integrated Flask REST APIs with Firebase Firestore and Firebase Storage for student enrollment, face embeddings, authentication, attendance logging, and report generation.
- Developed a web interface using HTML, CSS, and JavaScript with automated attendance marking, CSV reporting, and attendance analytics.

Air Quality Index Prediction using Regression Real-Time | Mini Project

- Collected and preprocessed real-time AQI data from 94+ Indian cities using OpenWeather and IQAir APIs.
- Trained and compared Linear Regression, Random Forest, and XGBoost models, achieving up to 89% accuracy.
- Automated real-time data collection and periodic model retraining using GitHub Actions and deployed the workflow on Render.
- Developed a prediction pipeline to forecast AQI for the next 1–48 hours and generate health-related alerts.

INTERNSHIP EXPERIENCE

WellDoc — QA Engineer Trainee **September 2026 – Present**

- Work as a QA Engineer Trainee on assigned projects, supporting software quality and validation activities throughout the testing lifecycle.
- Create and maintain detailed test cases based on project requirements and expected application behavior.
- Perform manual testing including functional, regression, smoke, and sanity testing for application features and enhancements.
- Identify, document, and track defects using Jira, including clear reproduction steps, expected results, actual results, and defect status.
- Use Confluence to refer to project documentation, requirements, testing information, and team knowledge resources.
- Perform defect retesting and regression validation after fixes to ensure functionality works as expected.
- Follow SDLC and STLC practices while collaborating with team members during project testing activities.

LEADERSHIP & ACHIEVEMENTS

Event Coordinator | TechNova 2025 & TechNova 2.0 2026 (IET Student Chapter)

- Planned and coordinated technical events for TechNova 2025 and TechNova 2.0 2026 in collaboration with the organizing team.

Student Placement Coordinator | Department of CSE (AI & ML)

- Coordinated placement-related activities between students and faculty
- Supported students during placement preparation and technical events.

CERTIFICATIONS & ACADEMIC HONORS

VTU BE Honors Programme – Visvesvaraya Technological University

- Programming in Java – 97%
- Introduction to Industry 4.0 and Industrial Internet of Things – 99%
- The Joy of Computing Using Python – 84%
- Introduction to Machine Learning – 85%
- Reinforcement Learning-100%
- Natural Language Processing-99%

NPTEL- Programming in Python (82%)